

tvPlayer as digital props for film and TV

This is basically a media player.

Power it and it will play the first image or video on the USB stick.

You can control the playback by keyboard or via a locally served website from your mobile device. Change aspect ratios, colors and playback speed, "change the channel"; now you have an interactive TV station and your actor or actress always watches what is in the script.

Fill a USB stick with video (or *image!*) files; plug it in; have a fake TV.

Channel number is shown when switching them, white noise is shown when nothing is available and the volume bar is there for you when you expect it. Both can be turned off on-the-fly.

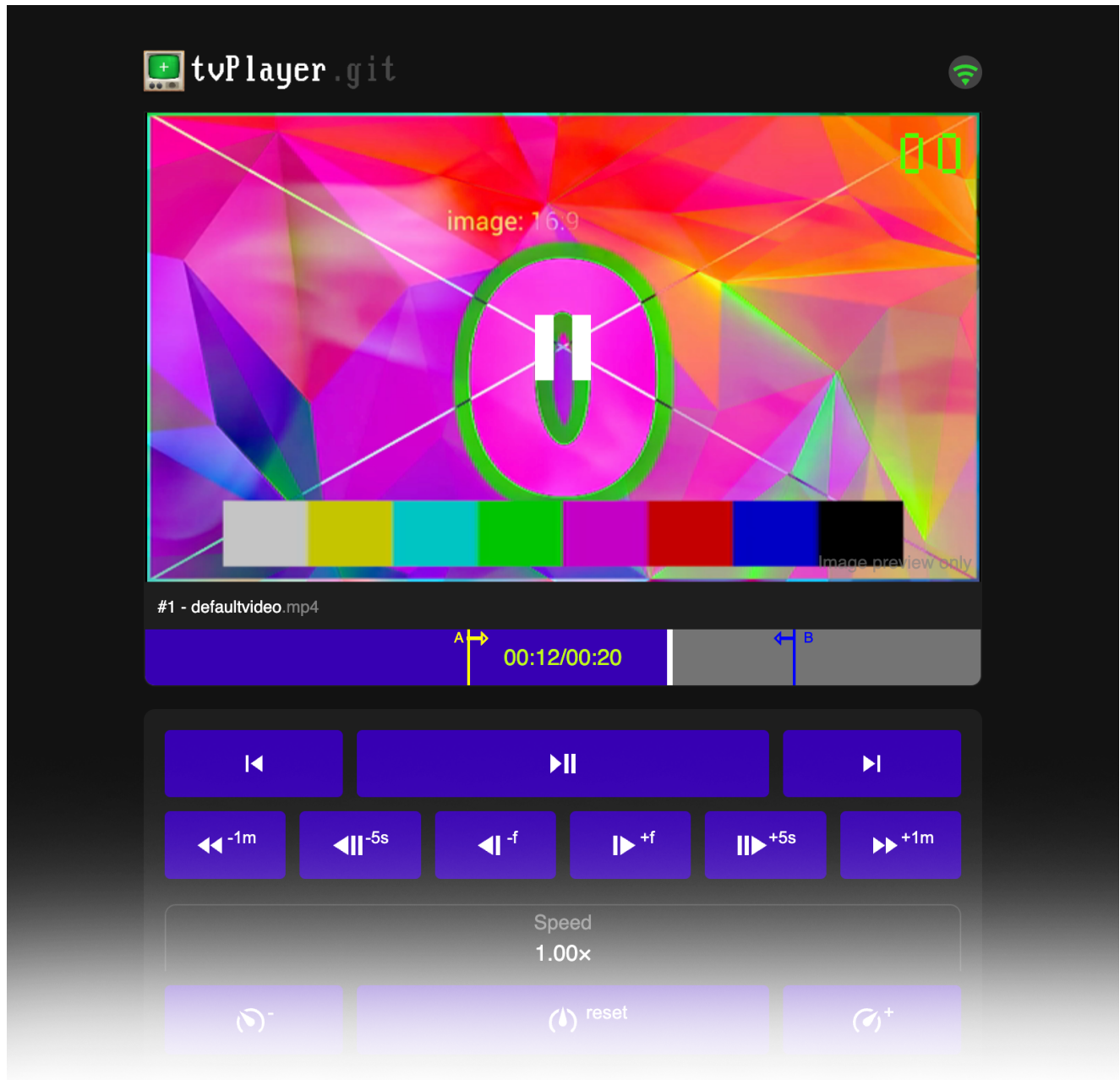
What you can do

- You can set inpoints for each video, so when you switch channels, it starts directly at the key moment, like when the journalist says, "*...some minor accident today in the hospital of Dunwich was contained and we have nothing to worry about. Now to the weather..*"
- You can turn the TV fake-off ("blackout").
- You can switch video-fitting-mode from contain to stretch to cover, to get rid of potential letterboxes.
- Control saturation/brightness/contrast
- Control playback speed
- Set inpoints on each video, so the video plays from that point whenever you select the video
- Loop: Set in- and outpoints to be looped
- Zoom and pan video
- You can turn off the fancy effects.
- The files do not auto-progress from one to another. Each video loops itself until you "switch the channel" (play the next file on the USB device).
- Change the brightness of the video to match exposure times of the camera.
- If everything's lost, there's a [green screen](#) button
- Control the playback with the webremote from your mobile device



Webremote

Once booted, connect your mobile device to the SSID **tvPlayer** and visit <http://10.3.141.1:8080>.



See full screenshot [here](#)

UDP

Send UDP commands to control the player:

```
echo -n "tvPlayer_go_to_channel:3" | nc -u -w1 192.168.1.153 53534
```

Whereas `tvPlayer` matches the hostname of the device.

`go_to_channel` is an example command. See the `handle_command()` function to see which commands are available.

`:3` is the value `3` for the command (optional; colon is separator).

`53534` is the default port.

Every accepted message is answered with `tvPlayer_acknowledged`, sent back to the **source port** of the sender.

Listen for it with `nc -u -l 53545` (or whichever port you sent from).

Sending UDP

`tvPlayer` can also send messages itself via `udp_send("tvPlayer_whatever")`.

Without arguments it sends to the default target configured at the top of `tvPlayer.py`:

```
UDP_LOCAL_PORT = 53534           # port this device listens on
UDP_TARGET_IP   = "192.168.1.100" # default target for udp_send()
UDP_TARGET_PORT = 53545           # default target port for udp_send()
```

A specific target can be given per message: `udp_send("tvPlayer_hello", "192.168.1.42", 5006)`.

The UDP logic lives in the reusable `udp.py` module (`UDPNode`).

Hardware

Tested successfully on Raspberry Pi 4 (Debian Bookworm 64-bit Lite).

Raspberry Pi 3 (Debian Bookworm 32bit) may **not** work.

Combine it with a [HDMI-to-RCA](#) adapter **and** a [RCA-to-RF](#) adapter to display on an old TV. Even sound works!

Get this up and running

1. Clone this repo `git clone https://github.com/falue/tvPlayer`
2. Follow the setup guide in [SETUP_LITE.md](#)
3. Insert USB, start watching TV.
4. Connect your mobile device to the SSID `tvPlayer` and visit `http://10.3.141.1:8080` or send UDP commands.

File handling

MPV player: Playable media

The media player [MPV \(doc wiki\)](#) used here can play pretty much everything - however, filename-endings are fixed to work with [.mp4](#), [.mkv](#), [.avi](#), [.mxf](#), [.m4v](#) and [.mov](#) (case insensitive).

It uses ffmpeg to decode, so the [list of playable media](#) is huge.
According to the mpv.io website:

File Formats: mpv supports a wide variety of media formats, including popular video files (e.g., MP4, MKV), audio codecs (e.g., AAC, MP3), and subtitles.

Some cherry picked examples:

- Containers: MP4, MKV, AVI, WebM, OGG, FLV, and more.
 - Video Codecs: H.264, HEVC, VP8, VP9, AV1, MPEG-4, MPEG-2, and others.
 - Audio Codecs: AAC, MP3, Vorbis, FLAC, Opus, AC3, and DTS.
- Subtitle Formats: SRT, ASS, SSA, VTT, and embedded subtitle tracks in containers like MKV.

Manually tested:

- ☒ [.mp4](#) MPEG-4 AAC, H264
- ☒ [.avi](#) MPEG-4 mp3
- ☒ [.mkv](#) h264, yuv420p
- ☒ [.mxf](#) mpeg2video (4:2:2), yuv422p
- ☒ [.mp4](#) HEVC, H265
- ☒ [.mov](#) H264
- ☒ [.mkv](#) "4k UHD" h264 yuv420p works (but stuttering on raspberry pi4 @8gb)

Also works with *images*!

Filetypes for images: [.jpg](#), [.jpeg](#), [.png](#), [.gif](#), [.tiff](#), [.bmp](#).

NOTICE: [.png](#) does **not** work with color mode: "*indexed colors*"!

USB drive format

Works with [FAT32](#) and [EXFAT](#) formatted USB drives.

Does not work with MAC formatted drives.

Filenames, TV channel order and AV channel

The order of the files on the USB device lead the order of the TV channels.

The alphabetically first file (case insensitive, numbers before letters) is the first channel, etc.

If you want to change the order, rename the files appropriately.

TIP: If any file name starts with [AV](#) (mp4 or any other) its always the channel "zero" and the channel number is shown as "**AV**".

TIP 2: Change the starting number - e.g. you want the first file to be channel 52 - change

`tv_channel_offset` in `settings.json`.

Make sure you set this when tvPlayer is not running!

Dont offset more than `amount of files + n <= 100`

Channel number and volume bar styling

If you want to change the appearance of the channel numbers or the volume bar, update the `.png` files in `assets/channel_numbers/` and `assets/volume_bars/`.

Images use PNG alpha transparency.

Note: For simplicity, keep the image sizes the same as they are hard coded. Otherwise you have to make changes to the script (change the parameters of both `display_image(...)` instances)

Update system

To update the system:

1. Download this file on your computer: <https://github.com/falue/tvPlayer/archive/refs/heads/main.zip>
2. Put it on any USB drive and stick it in the tvPlayer. Don't change the file name (`tvPlayer-main.zip`).
3. Reboot the tvPlayer.
4. check if update is applied on the webremote: On the very bottom, check if the "tvPlayer Version & Last updated" is updated.

Keyboard controls

Keypress	Action	Note
LEFT	prev channel	aka "previous file" on USB drive (alphabetically)
RIGHT	next channel	aka "next file" on USB drive (alphabetically)
[number]	go to channel nr [number]	
UP	jump +5 seconds	
CTRL + UP	jump +60 seconds	
SHIFT + UP	pause and jump one frame forwards	
DOWN	jump -5 seconds	
CTRL + DOWN	jump -60 seconds	
SHIFT + DOWN	pause and jump one frame backwards	

Keypress	Action	Note
p or space	toggle play / pause	
q	shutdown raspberry pi	
Q (SHIFT+q)	exit program	
b	toggle black screen on or off	Also pauses the current video when entering black screen, resumes video when exiting black screen
g	toggle green screen	Available chroma screens : Green, Blue, Black, Grey, White Each with 3 different tracking marker arrangements and one without.
G (SHIFT+g)	cycle through green screens →	
CTRL + g	cycle through green screens ←	
c	video fitting	"Contain", "stretch" or "cover"
i	set inpoint	Set where this video starts to play when changing channels
I (SHIFT+i)	clear inpoint on this video	
o	set outpoint	Set where this video ends (loops back to inpoint)
O (SHIFT+o)	clear outpoint on this video	
a	toggle tv GUI on or off	Show green channel numbers and volume bar
w	toggle white noise on or off between channel changes	
W (SHIFT+w)	cycle through 4 different white noise effects	Also activates white noise between channel change and shows current white noise selection
j	decrease playback speed	Min. 0.01x
k	normalize playback speed	
l	increase playback speed	Up to 3x
x	pan video to the right	
X (SHIFT+x)	pan video to the left	
y	pan video downwards	

Keypress	Action	Note
Y (SHIFT+y)	pan video upwards	
CTRL + x (or CTRL + y)	reset pan	
.	zoom out	
SHIFT + .	zoom in	
CTRL + .	zoom reset	
,	reduce video brightness by 5%	
SHIFT + ,	increase video brightness by 5%	
m	reduce video contrast by 2.5%	
SHIFT + m	increase video contrast by 2.5%	
n	reduce video saturation by 2.5%	
SHIFT + n	increase video saturation by 2.5%	
+	reduce volume by 10%	
-	increase volume by 10%	